

AUTOMATED VEHICLE CLASSIFICATION & COUNTING

AVC&C | Group G25 | University of Bradford

REFERENCES & BIBLIOGRAPHY

Complete Citation List | All Project Documents | Harvard Format

Reference Usage Map

The table below shows which document each reference appears in. All 36 references are cited across the six project documents submitted for COS6032-E.

#	Document Used In	Category	Reference
1	Research, Design, Testing	AI Model — Detection	Bochkovskiy et al. (2020) — YOLOv4
2	Research, Design, Testing, README	AI Model — Detection	Ultralytics (2024) — YOLO11n Docs
3	Research, Design, Testing	AI Model — Detection	Wang et al. (2023) — YOLOv9
4	Research, Design, Testing, README	AI Model — Tracking	Wojke et al. (2017) — DeepSORT
5	Research, Testing	Benchmark Dataset	Wen et al. (2020) — UA-DETRAC
6	Research, Testing	Benchmark Dataset	Lin et al. (2014) — COCO Dataset
7	Research	AI Model — Detection	Redmon et al. (2016) — YOLOv1
8	Research	AI Model — Detection	Ge et al. (2021) — YOLOX
9	Research	AI Model — Tracking	Zhang et al. (2022) — ByteTrack
10	Research, Design	AI Model — Detection	Ren et al. (2015) — Faster R-CNN
11	Research, Design	AI Model — Detection	Lyu et al. (2023) — RT-DETR
12	Research, Design	Dataset Platform	Roboflow (2024) — Platform Docs
13	Research, Design	Dataset	Geiger et al. (2012) — KITTI
14	Design, Testing	Core Library	Bradski (2000) — OpenCV
15	Design, Testing, README	Dashboard	Streamlit Inc. (2024) — Streamlit Docs
16	Design, Testing	Visualisation	Plotly Technologies (2015) — Plotly Python
17	Design, Testing	Data Processing	McKinney (2010) — Pandas
18	Ethics, GDPR, FAIR	Legal — Privacy	UK GDPR / Data Protection Act 2018
19	Ethics, FAIR	Governance	CDEI (2021) — AI Assurance Framework
20	Ethics	Policy — UK	DCMS (2021) — UK National AI Strategy
21	Ethics	Policy — Bradford	Bradford Council (2023) — Net Zero 2038
22	Ethics	Legal — Privacy	ICO (2021) — AI and Data Protection
23	Ethics	Policy — Transport	DfT (2022) — Local Transport Plan Guidance

24	Ethics	Policy — Bradford	Bradford Council (2023) — Digital Strategy
25	FAIR Report	AI Tool	Anthropic (2024) — Claude AI
26	Agile, Design	Methodology	Schwaber & Sutherland (2020) — Scrum Guide
27	Agile	Methodology	Beck et al. (2001) — Agile Manifesto
28	Design, Agile	Methodology	Crawley et al. (2007) — CDIO Framework
29	Research	Industry	Hikvision (2023) — ANPR Product Docs
30	Testing	Benchmark	Bewley et al. (2016) — SORT Algorithm
31	Research	Dataset	Cordts et al. (2016) — Cityscapes Dataset
32	Testing	Core Library	GitHub (2024) — Ultralytics Repository
33	Ethics	Legal	ICO (2023) — CCTV Code of Practice
34	Research	AI Model	Howard et al. (2017) — MobileNets
35	Ethics	Policy	Bradford Council (2022) — Local Plan
36	Design	Architecture	Kalman (1960) — Kalman Filter (DeepSORT basis)

A. AI Models & Algorithms

Object Detection — YOLO Family

- [1] Bochkovskiy, A., Wang, C.-Y. and Liao, H.-Y. M. (2020) *YOLOv4: Optimal Speed and Accuracy of Object Detection*. arXiv preprint arXiv:2004.10934. Available at: <https://arxiv.org/abs/2004.10934> [Accessed: April 2026]. Used in: Research Document (model comparison rationale)
- [2] Ultralytics (2024) *YOLO11 Technical Documentation*. Ultralytics Documentation. Available at: <https://docs.ultralytics.com> [Accessed: April 2026]. Used in: Research Document, Design Document, Testing & Validation Report, README
- [3] Wang, C.-Y., Yeh, I.-H. and Liao, H.-Y. M. (2023) *YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information*. arXiv preprint arXiv:2402.13616. Available at: <https://arxiv.org/abs/2402.13616> [Accessed: April 2026]. Used in: Research Document (model evaluation)
- [4] Redmon, J., Divvala, S., Girshick, R. and Farhadi, A. (2016) *You Only Look Once: Unified, Real-Time Object Detection*. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 779–788. doi: 10.1109/CVPR.2016.91. Used in: Research Document (foundational YOLO context)

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- [6] Ren, S., He, K., Girshick, R. and Sun, J. (2015) *Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks*. *Advances in Neural Information Processing Systems (NeurIPS)*, 28, pp. 91–99. doi: 10.1109/TPAMI.2016.2577031. Used in: Research Document, Design Document (rejected model)
- [7] Lyu, W. et al. (2023) *DETRs Beat YOLOs on Real-time Object Detection (RT-DETR)*. arXiv preprint arXiv:2304.08069. Available at: <https://arxiv.org/abs/2304.08069> [Accessed: April 2026]. Used in: Research Document (model comparison)
- [8] Howard, A. G., Zhu, M., Chen, B. et al. (2017) *MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications*. arXiv preprint arXiv:1704.04861. Available at: <https://arxiv.org/abs/1704.04861> [Accessed: April 2026]. Used in: Research Document (SSD MobileNet comparison)

Multi-Object Tracking

- [9] Wojke, N., Bewley, A. and Paulus, D. (2017) *Simple Online and Realtime Tracking with a Deep Association Metric*. *IEEE International Conference on Image Processing (ICIP)*, pp. 3645–3649. doi: 10.1109/ICIP.2017.8296962. Used in: Research Document, Design Document, Testing & Validation Report, README (core tracking algorithm)
- [10] Bewley, A., Ge, Z., Ott, L., Ramos, F. and Upcroft, B. (2016) *Simple Online and Realtime Tracking (SORT)*. *IEEE International Conference on Image Processing (ICIP)*, pp. 3464–3468. doi: 10.1109/ICIP.2016.7533003. Used in: Research Document (SORT vs DeepSORT comparison), Testing & Validation
- [11] Zhang, Y., Sun, P., Jiang, Y. et al. (2022) *ByteTrack: Multi-Object Tracking by Associating Every Detection Box*. *European Conference on Computer Vision (ECCV)*, *Lecture Notes in Computer Science*, vol. 13682. doi: 10.1007/978-3-031-20047-2_1. Used in: Research Document (tracking algorithm comparison)
- [12] Kalman, R. E. (1960) *A New Approach to Linear Filtering and Prediction Problems*. *Journal of Basic Engineering*, 82(1), pp. 35–45. doi: 10.1115/1.3662552. Used in: Design Document (Kalman filter — theoretical basis of DeepSORT tracking)

B. Datasets & Benchmarks

- [13] Wen, L., Du, D., Cai, Z. et al. (2020) *UA-DETRAC: A New Benchmark and Protocol for Multi-Object Detection and Tracking*. *Computer Vision and Image Understanding*, 193, p. 102907. doi: 10.1016/j.cviu.2020.102907. Used in: Research Document, Testing & Validation Report (primary external benchmark)
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C. Software Tools & Libraries

- [18] Bradski, G. (2000) *The OpenCV Library*. Dr. Dobb's Journal of Software Tools. Available at: <https://opencv.org> [Accessed: April 2026]. Used in: Design Document, Testing & Validation (video capture, frame processing, bounding box rendering)
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- [22] Ultralytics (2024) *Ultralytics GitHub Repository — YOLO11 Source Code*. GitHub. Available at: <https://github.com/ultralytics/ultralytics> [Accessed: April 2026]. Used in: Design Document, Testing & Validation (model weights, training API)
- [23] Anthropic (2024) *Claude AI — Large Language Model Assistant*. Anthropic. Available at: <https://claude.ai> [Accessed: April 2026]. Used in: FAIR Report (declared AI tool — code debugging and documentation guidance)

D. Legislation, Policy & Governance

Data Protection & Privacy

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- [27] Information Commissioner's Office (ICO) (2023) *CCTV and Surveillance Camera Code of Practice*. ICO. Available at: <https://ico.org.uk/media/for-organisations/documents/1542/cctv-code-of-practice.pdf> [Accessed: April 2026]. Used in: Ethics & Implications (public surveillance compliance)

AI Ethics & Governance

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Bradford Council & Transport Policy

- [30] Bradford Council (2023) *Bradford District Net Zero 2038 Action Plan*. Bradford Metropolitan District Council. Available at: <https://www.bradford.gov.uk/climate-change> [Accessed: April 2026]. Used in: Ethics & Implications (sustainability alignment)
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E. Project Methodology

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- [36] Crawley, E., Cameron, B. and Selva, D. (2007) *System Architecture: Strategy and Product Development for Complex Systems — CDIO Framework*. Prentice Hall. ISBN: 978-0135783023. Used in: All project documents (CDIO Conceive-Design-Implement-Operate lifecycle)

F. Industry References

- [37] Hikvision UK (2023) *ANPR and Traffic Counting Camera Systems — Product Documentation*. Hikvision. Available at: <https://www.hikvision.com/uk> [Accessed: April 2026]. Used in: Research Document (commercial benchmark pricing context)

Reference Count Summary

Category	References	Ref Numbers
AI Models & Algorithms	12 references	[1]–[12]

Datasets & Benchmarks	5 references	[13]–[17]
Software Tools & Libraries	6 references	[18]–[23]
Legislation, Policy & Governance	10 references	[24]–[33]
Project Methodology	3 references	[34]–[36]
Industry References	1 reference	[37]
TOTAL	37 references	[1]–[37]